

## Overview

Welcome to **MIE 1626 – Data Science Methods!** The course introduces core techniques in data science, in lectures, tutorials, and mini-projects. The lectures, tutorials, and mini-projects will support students' work on their course project and on future research and professional work involving data. Students will select a dataset of interest to them and produce an analysis and a project report. Students will use domain knowledge and technical expertise to produce their analyses.

## Website &amp; Forum

Website: <http://www.cs.toronto.edu/~guerzhoy/1626s26>

Forum: <https://piazza.com/utoronto.ca/summer2026/mie1626>

All course handouts will be posted on the course website. *Students are responsible for reading all announcements on the course forum on Piazza.*

## References

There is no required textbook. Lecture slides will be posted on the course website. The following are recommended as references.

- Andrew Gelman and Jennifer Hill. *Data Analysis using Regression and Multilevel/Hierarchical Models*. Cambridge University Press, 2006.
- Cosma Rohilla Shalizi. *Advanced Data Analysis from an Elementary Point of View*. Cambridge University Press (forthcoming). Preprint available at <https://www.stat.cmu.edu/~cshalizi/ADAfaEPoV/>
- Fred Ramsey and Daniel Schafer. *The Statistical Sleuth: A Course in Methods of Data Analysis*, 3rd ed. Brooks/Cole, 2013.
- Kieran Healy. *Data Visualization: A Practical Introduction*. Princeton University Press, 2026.

## Instructor

| Instructor       | Email                                                                | Office  | Office Hours   |
|------------------|----------------------------------------------------------------------|---------|----------------|
| Michael Guerzhoy | <a href="mailto:guerzhoy@cs.toronto.edu">guerzhoy@cs.toronto.edu</a> | BA 2028 | By appointment |

## Contact

Lectures and tutorials take place on Wednesdays and Fridays, 2:00pm–5:00pm. Students are expected to attend both lecture and tutorial blocks.

## Grading

The grading scheme and schedule for the course is as follows.

|                                                | Worth | Due                       |
|------------------------------------------------|-------|---------------------------|
| Mini-Project 1                                 | 5%    | May 16                    |
| Preliminary Project Proposal                   | 2%    | May 19                    |
| Mini-Project 2                                 | 5%    | <b>May 28</b>             |
| Test 1                                         | 10%   | <b>May 29</b>             |
| Mini-Project 3                                 | 5%    | <b>July 25</b>            |
| Mini-Project 4                                 | 5%    | <b>July 25</b>            |
| Preliminary Project Presentation<br>+ Document | 8%    | Jun 15                    |
| Test 2                                         | 30%   | <b>Wed Aug 12, 2:00pm</b> |
| Course Project                                 | 30%   | <b>Aug 10</b>             |

## Late work

Each student starts the term with 7 “grace days.” You can use grace days to submit late work with no penalty. For example, a student who submits their work 25 hours after the deadline will use up two grace days. You cannot use more than 3 grace days at a time. Grace days do not apply to the preliminary project presentation, tests, or the final project.

**Course  
project**

Each student (or pair of students) in the course will work on a large-scale data science project. Students are encouraged to pick a dataset of interest to them. Students should aim to achieve one or several of the following.

- Obtain interesting insights about the dataset
- Apply interesting data analysis techniques to the dataset
- Demonstrate a new data analysis technique and apply it to the dataset
- Collect a novel dataset

Course projects will contain

- A description of the dataset being analyzed, and a summary of the relevant domain knowledge. For example, if working with audio data, a summary of how sound is generated, perceived, and processed would be appropriate. If working with hospital data involving length-of-stay times, it would be appropriate to summarize the relevant literature on hospital lengths of stay.
- A summary of previous work that is related to the data analysis project.
- An exploratory analysis of the dataset being analyzed.
- A description of the method being applied, and a summary of related work.
- A description of the outcomes of the analysis, or an analysis of a predictive system that was built.

Students will submit a preliminary project proposal by May 19 and a preliminary project document, which should contain a full description of the project and initial exploratory data analysis results, by Jun 15. Each student will give a preliminary project presentation to the class on Jun 15.